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Forecasting presidential elections: Accuracy of ANES voter intentions

Hyein Ko [a,*], Natalie Jackson [b], Tracy Osborn [c], Michael S. Lewis-Beck [c]

[a] Department of Political Science, University of Illinois Urbana-Champaign, Urbana, IL, USA
[b] GQR Research, Washington, DC, USA
[c] Department of Political Science, University of Iowa, Iowa City, IA, USA

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ABSTRACT

Despite research on the accuracy of polls as tools for forecasting presidential elections, we lack an assessment of how accurately the ANES, arguably the most used survey in political science, measures aggregate vote intention relative to the actual election results. Our ANES 1952–2020 results indicate that the reported vote from the post-election surveys accurately measures the actual vote (e.g., it is off by 2.23 percentage points, on average). Moreover, the intended vote measure from the pre-election surveys reasonably accurately predicts the actual aggregate popular vote outcome. While outliers may exist, they do not appear to come from variations in the survey mode, sample weights, time, political party, or turnout. We conclude that political scientists can confidently use the intended vote measure, keeping in mind that forecasting the popular vote may not always reveal the actual winner.

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1. Introduction

The American National Election Studies (ANES) is commonly referred to as the gold standard survey for scientific research on American voting behavior. They have been the source of several noteworthy scholarly volumes, not to mention an abundance of journal articles (Campbell et al., 1960; Fiorina, 1981; Goren, 2013; Lau & Redlawsk, 1997; Lewis-Beck et al., 2008; Petrocik, 2009). Besides academics, leading politicians themselves use them. An outstanding example comes from candidate Hillary Clinton when, in her memoir, What Happened, she called on this "gold standard" survey to help explain defeat in the 2016 presidential contest (Clinton 2017, p. 412). Her hypotheses about loss were submitted to formal test against the 2016 ANES data by at least two teams of political science scholars (Lewis-Beck and Quinlan 2019; Stewart et al. 2019).

Of course, the gold standard argument for the ANES has not gone unchallenged. Aldrich (2017, p. 10), in a Board Report to the ANES itself, notes its rivalry with the General Social Survey (GSS) and goes on to point out that it does not always meet its 'gold standard' goals. In his words, "conditions that underpinned this level of success have eroded". For example, there are problems with rising non-response rates, difficulties with the face-to-face survey mode, departures from canons of probability sampling, and improper construction of the sampling frame. A nagging question is the identification of actual voters (Bernstein et al., 2001; Jackman & Spahn, 2019; Traugott, 1989). Also, we have ongoing concerns about measurement in the vote intention item itself (Achen & Blais, 2015). Campbell (2010, pp. 616–617), in a detailed analysis of ANES voting data (1952–2008) points out various inaccuracies, including the 2008 results, which gave Barack Obama 54.8 percent of the two-party popular vote and projected a turnout of 77.6 percent (when in fact the

* Correspondence to: Department of Political Science, University of Illinois Urbana-Champaign, 431 David Kinley Hall, Urbana, IL 61801, USA.
E-mail addresses: hyeink@illinois.edu (H. Ko), nmjb09@gmail.com (N. Jackson), tracy-osborn@uiowa.edu (T. Osborn), michael-lewis-beck@uiowa.edu (M.S. Lewis-Beck).

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Table 1
ANES variable description.

ANES variable	Description
VCF0004	Year of study (4-digit)
VCF0006	Study respondent number: year-level case ID
VCF0702	Did the respondent vote in the national elections?
VCF0704a	Vote for president – Major parties
VCF0713	President pre-election intent for a vote for the president
VCF0009z	Weight: For 1970 type 0 – 2012, 2016, 2020 full sample

more accurate numbers were, respectively, 53.7 percent and 61.7 percent). He observes that "Despite inaccuracies of this sort ... NES data are routinely reported as what the electorate thinks and does".

Here, we explore the accuracies and inaccuracies of ANES presidential election data from 1952 to 2020 with respect to reported votes and, more importantly, with respect to intended votes. We employ the aggregate two-party popular vote for president as a benchmark for evaluating these measures. First, we compare the aggregated reported vote to the aggregated actual vote. We then assess how well the vote intention question of the regular pre-election ANES survey, carried out about one to two months before the contest, forecasts the actual vote outcome. While methodological, this contribution adds value to the burgeoning American national election forecasting literature. In the political science community, scientific presidential election forecasts have come to gain notice partly due to visible professional symposia (Campbell, 2016; Campbell & Lewis-Beck, 2008; Linzer & Lewis-Beck, 2015). A recent example comes from PS: Political Science and Politics, which sponsored a symposium forecasting the 2020 presidential elections (Dassonneville & Tien, 2021). In these and related efforts, the systematic use of the ANES aggregate vote intention item as a forecasting instrument has yet to be pursued.

Survey researchers often say that such pre-election polls are "snapshots", meaning that they show what the situation looks like at the time the survey is fielded; they are not intended to forecast the election outcome. However, polls of this sort have frequently been used as forecasting tools in popular media as well as within academic forecasts, particularly since 2008 (Jackson, 2018). ANES is typically not included in these exercises or the later analyses of the accuracy of these polls due to the data and results being released months after the election. Unlike many pre-election horse race polls, ANES is not primarily designed to estimate who the most likely voters are; instead, it is intended to be the best possible sample of all Americans in the population, regardless of whether they vote or not. At the same time, ANES continues to be hugely important in explaining electoral politics in the political science literature, and it commands a large investment of money from the National Science Foundation, providing a dataset many scholars continue to use. These considerations warrant investigating how it performs on key outcomes the survey seeks to measure. Given the voter turnout question is known to be flawed (Jackman & Spahn, 2019), we should investigate how reliable vote choice is as an independent or dependent variable.

Our analysis proceeds in stages. First, we assess the error in the post-election reported aggregate vote. Second, we evaluate errors in the pre-election intended aggregate vote. Next, we explore the determinants of these errors, including the possible impact of sample weighting, interview mode, party incumbency, political epoch, and turnout. We compare ANES to other polls in predictive ability and then offer conclusions regarding the quality of ANES aggregate vote intention as a forecasting tool for American presidential election outcomes.

2. Description of data and variables

Our main data are gathered from the ANES (2022), particularly the time series cumulative data,1948–2020, released on September 16, 2022. While questionnaires were differently stated in a few years, ANES repeatedly asked (1) whether the respondent voted for the president in terms of a major party and (2) whether the respondent intended to cast their votes for the upcoming presidential election. Table 1 describes the six key variables we use in our analysis. The first variable is *VCF0004*, which indicates the year of study, ranging from 1948 to 2020. *VCF0006* provides the identification information for each respondent in the given year of study. *VCF0702* indicates whether a respondent voted in the national elections, although the exact questionnaires varied depending on the year of study. The fourth variable is *VCF0704a*, which asked respondents who they voted for in the Presidential election between Democrat and Republican party candidates. The fifth variable, *VCF0713*, is about whom respondents intended to vote for in the presidential election; this question was asked during the pre-election study period. Lastly, *VCF0009z* is the weights variable. In the 1970 ANES study, ANES designed two questionnaire forms to obtain maximum information from Black respondents while reducing survey duration for white respondents. This created Type 0 (asked to every participant), Type 1 (a set of race-related questions excluded from the Type 1 assigned white respondents), and Type 2 (a set of questions on various other topics excluded from the Type 2 assigned white respondents). Due to this unique format, the 1970 study uses variable-level weighting. ANES codebook notes whether a researcher should consider specific weights based on the questionnaire; vote-related variables for our study have VCF0009x, VCF0009y, or VCF0009z as options. We apply VCF0009z because we use the combined sample for our analyses, while x is for the FTF-only sample, and y is for the Web-only sample.

In addition to ANES, we gathered election-related data from different sources. Actual two-party election data are obtained from Campbell (2010) Table 1 and MIT Election Data and Science Lab (2017). Actual national turnout

Table 2
Reported and actual two-party vote for presidential candidates (1952–2020).

Democratic presidential two-party vote percentage					
Election	Actual	Unweighted		Weighted	
		Reported	Difference (reported - actual)	Reported	Difference (reported - actual)
1952*	44.60	41.94	−2.66	41.94	−2.66
1956*	42.25	40.36	−1.89	40.36	−1.89
1960	50.08	49.67	−0.41	49.19	−0.89
1964*	61.34	67.51	6.17	67.51	6.17
1968*	49.60	46.21	−3.39	46.21	−3.39
1972*	38.21	35.66	−2.55	35.66	−2.55
1976	51.14	50.68	−0.46	51.15	0.01
1980*	44.84	43.67	−1.17	43.67	−1.17
1984*	40.88	41.79	0.91	41.79	0.91
1988*	46.17	47.11	0.95	47.11	0.95
1992	53.62	58.44	4.82	58.50	4.88
1996	54.80	58.03	3.22	58.44	3.64
2000	50.26	52.68	2.42	52.01	1.75
2004	48.77	49.20	0.43	49.90	1.13
2008	53.77	66.60	12.83	54.83	1.06
2012	52.01	59.60	7.59	53.61	1.60
2016	51.16	52.28	1.12	52.95	1.79
2020	52.27	57.80	5.54	55.99	3.72
Mean absolute error of reported presidential vote percentage (unweighted)					± 3.25
Mean absolute error of reported presidential vote percentage (weighted)					± 2.23

Note: From 1952–1972, actual two-party vote data is obtained from Campbell (2010) Table 1. From 1976 to 2020, the table uses MIT Election Data and Science Lab (2017) for actual two-party vote.
*: Survey years with no weighting adjustment. Several years lack weights from ANES data, leading to the inconsistency between weighted and unweighted years.

rates data comes from McDonald (2023). Lastly, presidential election poll data is gathered from Campbell (1996), Gallup (2023), RealClearPolitics (2023a, 2023b), Woolley and Peters (2023).

3. Reported vote errors, post-election: ANES data

As a first step in investigating this issue, we ask a simple question: When we eyeball the data for the 1952–2020 presidential elections, do large errors jump out in the post-election reported vote choice? Table 2 shows the reported vote and actual vote for the Democratic candidate (using the two-party popular vote percentage) across the period. We focus on the weighted data since we expect the situation to improve when available weights are imposed to adjust for differential socio-demographic response rates, and indeed, it does. In Table 2, the difference between reported and actual vote percentage (column 6) ranges from −3.39 to 6.17. (This last value, 6.17, is for 1964 and stands out from the other errors). Over time, a clear trend has emerged, with the differences from 1984 onward always being positively signed, while most of the differences from 1980 backward are negatively signed, suggesting a contemporary bias overestimating Democratic candidates. We return to this point later.

These raw ANES estimates of the actual vote from the reported vote appear reasonably accurate, as the Mean Absolute Error (MAE) of just 2.23 implies. However, a strong statistical and theoretical defense of this evaluation is important. First, with respect to statistical theory, we assume there will be sampling error (SE) because we are estimating a population value (a vote share). Conventionally, the standard sampling error calculation assumes that the sample is a probability sample, represented by a simple random sample. In practice, for national public opinion surveys such as the ANES, the sample generally follows some form of multistage cluster sample with a non-zero probability selection of units. These complexities, which often seem necessary, tend to somewhat increase the estimate of the SE. For example, suppose the goal is estimating the percentage of two-party support for the Democratic party, based on such a sample survey at the University of Michigan (SRC), sample size = 1000. If it were a simple random sample, the expected sampling error would = 3.2. Thus, if the vote estimate were 45 percent, then that would suggest that in the population, the true estimate was 45 (± 3.2) for a 95 percent confidence interval (CI). If, however, the sample were a stratified cluster sample done with probability methods, the confidence interval would increase somewhat at 45 (± 4). We see that the CI is wider, suggesting a wider band of error. If we wanted to reduce the error by doubling the sample size to 2000, that would decrease the sampling error by one point, respectively. This suggests that an increasing sample size reduces the error; however, this reduction gets more and more costly in terms of power and money. Since most US elections can be correctly predicted within three or four percentage points, an $N = 1500$ for national public opinion surveys is not uncommon. These conditions suggest a frequently used standard for assessing accuracy: is the sample estimate four or more points from the actual result? (Weisberg and Bowen 1977, pp. 37–41; Bowen and Weisberg 1980, pp. 14–16). Applying this rule to the reported vote errors (see Table 2), only two estimates, 1964 and 1996, fall beyond that boundary. In other words, 16 of the 18 estimates would fall within the expected sampling error.

Table 3
Intended and actual two-party vote for presidential candidates (1952–2020).

Democratic presidential two-party vote percentage					
Election	Actual	Unweighted		Weighted	
		Intended	Difference (intended - actual)	Intended	Difference (intended - actual)
1952*	44.60	44.18	−0.42	44.18	−0.42
1956*	42.25	42.56	0.31	42.56	0.31
1960	50.08	46.88	−3.20	45.42	−4.66
1964*	61.34	72.07	10.73	72.07	10.73
1968*	49.60	45.29	−4.31	45.29	−4.31
1972*	38.21	31.59	−6.62	31.59	−6.62
1976	51.14	50.37	−0.76	51.09	−0.05
1980*	44.84	50.84	6.00	50.84	6.00
1984*	40.88	41.21	0.34	41.21	0.34
1988*	46.17	47.75	1.59	47.75	1.59
1992	53.62	59.50	5.88	59.50	5.87
1996	54.80	62.83	8.02	63.46	8.66
2000	50.26	55.06	4.80	54.19	3.93
2004	48.77	51.57	2.81	51.94	3.17
2008	53.77	66.80	13.03	55.61	1.84
2012	52.01	60.31	8.30	54.28	2.27
2016	51.16	53.64	2.48	54.46	3.30
2020	52.27	55.48	3.22	54.49	2.22
Mean absolute error of intended presidential vote percentage (unweighted)					± 4.60
Mean absolute error of intended presidential vote percentage (weighted)					± 3.68

Note: From 1952–1972, actual two-party vote data is obtained from Campbell (2010) Table 1. From 1976 to 2020, the table uses MIT Election Data and Science Lab (2017) for actual two-party vote.
∗: Survey years with no weighting adjustment. Several years lack weights from ANES data, leading to the inconsistency between weighted and unweighted years.

4. Intended vote errors, pre-election: ANES data

In Table 3, we replicate Table 2, but for the intended vote measure in the pre-election surveys. Again, we focus on the weighted results, where seven elections exceed an absolute error value of 4.00. However, five are early in the series, i.e., 1980 or before. Overall, the MAE of these estimates registers 3.68, which is not unreasonable given the surveys are several weeks before the election. More importantly, for forecasting purposes, the winner of the vote (i.e., greater than 50 percent of the two-party popular vote) is correctly predicted 15 out of 18 times. That is, vote intention share and actual vote share point to the same winner (except in 1960, 1980, and 2004). Thus, we might expect the aggregate intended vote to forecast the winner of the popular presidential vote 83 percent of the time (i.e., $15/18 \approx .83$).

5. Vote choice errors and turnout errors

An obvious concern revolves around whether incorrect turnout estimates affect the accuracy of the intended vote. The ANES consistently overestimates voter turnout for several reasons. According to Jackman and Spahn (2019), the leading cause of overestimation is over-reporting, where respondents claim they voted but did not. Other significant causes include non-response bias, where those who choose to participate in the ANES are more likely to turn out than those who do not answer the survey. Jackman and Spahn (2019) estimate that these effects account for up to 13 percent of the reported turnout in the 2012 ANES face-to-face sample. Enamorado and Imai (2019) agree that overreporting is the primary source of overestimation. Both studies suggest validating turnout with voter file data; Fig. 1 shows actual and reported voter turnout from 1952 to 2020. We observe, as expected, that reported turnout consistently exceeds actual turnout. We can also see that the error has notable swings over time. In Table 4 (column 6), we estimate the amount of error by election. Our estimates are similar to those of Campbell (2010, p. 620). Here, we see a low of four points (in 1952) to a high of 14 points (in 2012). Overall, the MAE registers about eight percentage points.

A logical follow-up question addresses whether turnout overestimation affects our ability to estimate candidate preferences using the vote intention measure. It might do so if specific groups of people tend to overreport systematically. For instance, more educated and politically interested people tend to overreport turnout (Enamorado & Imai, 2019). Partisans tend to overreport in recent ANES and Cooperative Election Study (CES) surveys (Ansolabehere & Hersh, 2012). To assess the impact of this turnout overestimation on the accuracy of reported candidate preferences, we regress voter turnout differences on the two-party vote difference in Table 5. We observe that the voter turnout variable lacks statistical significance at conventional levels. Further, the statistical fit of the model is quite small, i.e., the adjusted R-squared is only .06. Thus, we conclude that while voter turnout overestimation suggests a problem, it does not significantly perturb the efficacy of the election forecasts using the ANES vote intention item as a predictor. However, besides the role of turnout, we need to explore other possible systematic influences on such presidential election forecasts.

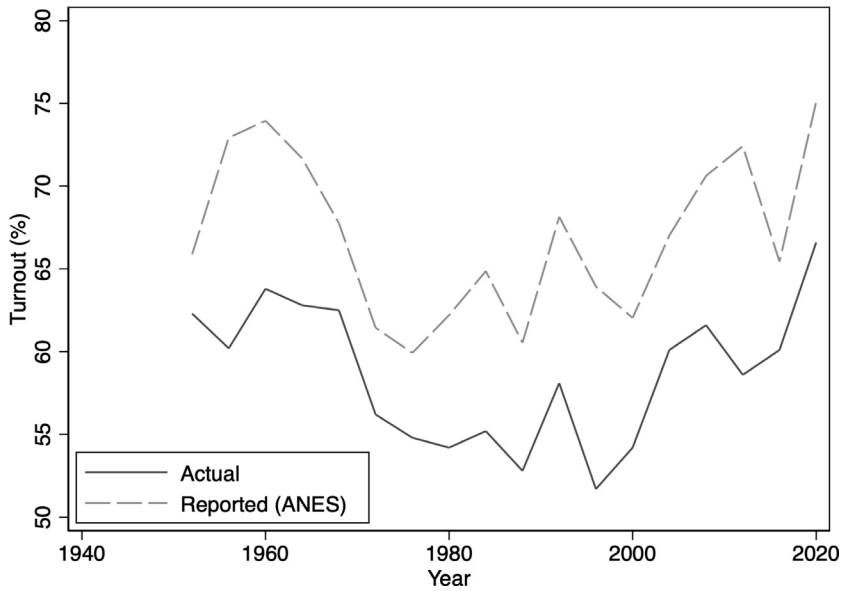


Fig. 1. Estimated turnout (ANES) and actual turnout (1952–2020), weighted.

Table 4
Estimated turnout (ANES) and actual turnout (1952–2020).

Election	Actual	Unweighted		Weighted	
		Estimated	Difference (estimated- actual)	Estimated	Difference (estimated - actual)
1952*	62.3	65.9	3.6	65.9	3.6
1956*	60.2	72.9	12.7	72.9	12.7
1960	63.8	77.2	13.4	74.0	10.2
1964*	62.8	71.7	8.9	71.7	8.9
1968*	62.5	67.8	5.3	67.8	5.3
1972*	56.2	61.4	5.2	61.4	5.2
1976	54.8	61.9	7.1	59.9	5.1
1980*	54.2	62.2	8.0	62.2	8.0
1984*	55.2	64.9	9.7	64.9	9.7
1988*	52.8	60.5	7.7	60.5	7.7
1992	58.1	68.4	10.3	68.1	10.0
1996	51.7	68.6	16.9	63.9	12.2
2000	54.2	65.4	11.2	62.0	7.8
2004	60.1	69.1	9.0	67.0	6.9
2008	61.6	69.0	7.4	70.6	9.0
2012	58.6	74.5	15.9	72.4	13.8
2016	60.1	67.6	7.5	65.4	5.3
2020	66.6	77.9	11.3	75.1	8.5
Mean absolute error of estimated turnout (unweighted)					± 9.5
Mean absolute error of estimated turnout (weighted)					± 8.3

Note: Voter turnout data comes from McDonald (2023).
∗: Survey years with no weighting adjustment. Several years lack weights from ANES data, leading to the inconsistency between weighted and unweighted years.

6. Predictors of the two-party vote share: Multivariate analysis

Table 6 presents different regression (OLS) models for predicting the actual two-party popular vote over the time series. Model 1 offers our core equation for forecasting the aggregate two-party popular vote from the aggregate intended vote (AIV). The model fit, Adjusted R-squared = .85, is high, as expected. Model 2, with an aggregate reported vote as the predictor, yields a still higher fit, which is unsurprising since it is measured ex-post facto. The other models test the influence of different possible covariates. Model 3 fails to uncover a time trend effect, suggesting the actual vote itself has not become more difficult to forecast, a finding that accords with Nadeau et al. (2020). Model 4 shows a statistically negative effect of weighting, provocatively suggesting some decline in accuracy when the presence of weights alone is used as a predictor. However, the magnitude of the effect is quite small and substantively inconsequential. Model 5, encouragingly, shows no differential impact due to the choice of mode, face-to-face or not. Similarly, Model 6

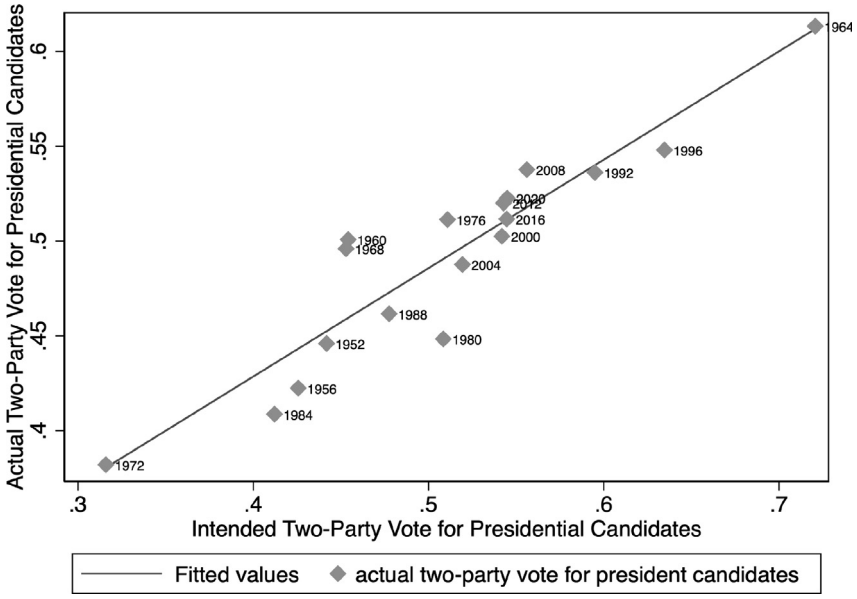


Fig. 2. Aggregated vote regressed on aggregated intended vote (1952–2020), weighted.

Table 5
Impact of voter turnout differences on the two-party vote differences, weighted.

	Model 1
Voter Turnout Difference	0.537
	(0.366)
Constant	−2.580
	(3.212)
N	18
R-Squared	0.119
Adj R-Squared	0.064
Root MSE	4.281

Note: Standard errors in parentheses. * $p < .05, **p < .01, ***p < .001$.

shows that the political party of the incumbent, Democrat or Republican, does not impact the accuracy of predicting the winner. Model 7, a multivariate specification, suggests that once the prime predictor variable of the intended vote is considered, the presence of weights lacks significance. Model 8 still has more control variables, and none attain statistical significance, indicating that the aggregate intended vote (AIV) variable offers a sound specification when the goal is prediction.

7. Regression diagnostics: Sample size and outliers

The estimates of Table 6 underline the promise of ANES as a toolbox for forecasting presidential elections. However, the analysis faces problems of a small N and apparent outlying observations (Lewis-Beck and Lewis-Beck 2015, p. 28). To check for these issues, first, we apply the comprehensive Shapiro–Wilk test for normality (see Table 7), which shows the hypothesis of normality cannot be rejected (at .05). While that result provides some encouragement, it rests in part on the small sample size. In other words, the possibility of normality does not mean that the data are free of outlier problems. Below, we explore the presence of outliers and their forecasting consequences.

A scatter plot with aggregate actual vote regressed on aggregate intended vote appears in Fig. 2. The regression line generally fits the observations snugly, as the adjusted R-squared of .85 would suggest, supporting the overall linearity assumption (recall Table 6 for this and other relevant statistics). Furthermore, the average out-of-sample prediction error is only a bit over two percentage points, as the root mean squared error (RMSE) suggests. Of course, within the sample, some observations appear to stray rather far from the line, e.g., 1960 and 1968, so we go on to more systematic outlier checks.

For our aggregated intended vote (AIV) model, 1960 and 1980 might be considered outliers, as they have studentized residuals greater than |2|. However, an overall "mean shift" jackknife test does not support the notion that 1960 and 1980 have statistically significantly different patterns from other observations, as Table 8 shows. Other tests explore whether deviating observations influence the magnitude of the AVI coefficient itself. A leverage test (see Table 9) suggests the 1964 and 1972 observations may have more influence on that coefficient than the other observations. This lacks surprise, considering that they had actual point prediction errors of 10.73 and −6.62. However, the dominant diagnostic for assessing the net influence of observations remains Cook's Distance, D (see Table 9). By that measure, the data reveal no suspect outliers, an unexpected but encouraging result (Anderson 2004, p. 940). If the Cook's D test stands accepted as a court of last resort, then worries are lessened.

Table 6
Determinants of actual vote, weighted.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Intended Vote	0.572***						0.528***	0.556***
	(0.058)						(0.060)	(0.069)
Reported Vote		0.701***						
		(0.041)						
Year			0.001					−0.001
			(0.001)					(0.000)
Weighted Impact				−0.058*			−0.018	−0.030
				(0.023)			(0.011)	(0.014)
Face-to-Face Survey					−0.031			−0.020
					(0.036)			(0.018)
Incumbent President's Party						0.034		−0.004
						(0.026)		(0.012)
Constant	0.200***	0.141***	−1.212	0.518***	0.518***	0.477***	0.230***	1.649
	(0.030)	(0.021)	(1.232)	(0.016)	(0.033)	(0.017)	(0.034)	(0.760)
N	18	18	18	18	18	18	18	18
R-Squared	0.859	0.948	0.107	0.280	0.046	0.094	0.882	0.909
Adj R-Squared	0.850	0.945	0.051	0.235	−0.014	0.037	0.866	0.871
Root MSE	0.022	0.013	0.055	0.049	0.056	0.055	0.021	0.020

Note: Standard errors in parentheses. * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 7
Shapiro–Wilk normality test for the residuals.

	Weighted
Observation	18
W	0.978
p-value	0.926

Note: Shapiro–Wilk test or -swilk- in Stata is used for a normality test on the following equation: Actual vote = b0 + b1*Intended vote + error. The null hypothesis is that the residuals are normally distributed. *W* is the test statistic.

Table 8
Mean shift test.

	Weighted
Outlier(s)	−0.001
	(0.017)
Intended Vote	0.572***
	(0.060)
Constant	0.200***
	(0.032)
N	18
R-Squared	0.859
Adj R-Squared	0.840
Root MSE	0.022

Note: Standard errors in parentheses. * $p < .05$, ** $p < .01$, *** $p < .001$.
Outliers are indicated when studentized residuals are greater than |2|. 1960 and 1980 are indicated outliers for this model.

Table 9
Leverage test and Cook's distance test.

Year	Leverage	Cook's D
1952	0.090	0.005
1956	0.108	0.062
1960	0.080	0.167
1964	**0.369**	0.002
1968	0.080	0.138
1972	**0.327**	0.002
1976	0.056	0.025
1980	0.056	0.118
1984	0.125	0.124
1988	0.064	0.010
1992	0.106	0.002
1996	0.164	0.054
2000	0.062	0.004
2004	0.056	0.006
2008	0.070	0.034
2012	0.066	0.007
2016	0.064	0.000
2020	0.064	0.010

Note: Both test was conducted with the following: *Actual vote* $= b0 + b1 * Intended\ Vote + error$. For the leverage test, two **bolded** observations (years) show greater leverage than the $(2k+2)/n$ criterion, where k is the number of predictors and n is the number of observations. The conventional cutoff $(4/n)$ is used for the Cooks' D, where n is the number of observations. No observation (year) has a greater value than this cutoff.

While concerns about outliers may remain, they are not necessarily fatal.

8. Comparing ANES 'Forecasts' to other 'Final' polls

One useful baseline for quality evaluation of forecast instruments built from ANES vote intention data is compared to commercial public opinion polls, such as the historic Gallup final trial-heat polls. Campbell (1996, p. 410–413) systematically carried out this task, looking at the Gallup series from 1948 to 1992. To quote Campbell (1996, p. 412), "correctly predicting the winning presidential candidate is ultimately the most important political test of a presidential forecasting model". He found that Gallup, with its early September poll, correctly foretold the winner in 10 out of 12 races. Following that strategy, we collected data on the same sort of early September measure, first from Gallup and, for 2016 and 2020, from other available surveys. The results appear in Table 10.

Table 10
The accuracy of polls in early september of the election year, 1952–2020.

Election	Early september poll for incumbent party's candidate	Incumbent presidential party two-party vote (%)	Absolute percentage point error of poll	Predicted winner?
1952	42.5	44.63	2.13	Right
1956	55.5	57.70	2.20	Right
1960	50.5	49.92	0.59	Wrong
1964	68.0	61.29	6.71	Right
1968	44.0	49.65	5.65	Right
1972	62.5	61.58	0.92	Right
1976	41.0	48.95	7.95	Right
1980	48.8	45.15	3.65	Right
1984	59.5	59.11	0.40	Right
1988	54.0	53.90	0.10	Right
1992	42.5	47.25	4.75	Right
1996	55.5	54.80	1.80	Right
2000	48.5	50.26	3.26	Wrong
2004	52.0	51.23	0.77	Right
2008	45.0	46.23	1.23	Right
2012	49.0	52.01	3.01	Wrong
2016	44.0	51.16	7.16	Wrong
2020	42.0	47.73	5.73	Right
		Mean error	± 3.22	14 Right
		Median error	± 2.671	4 Wrong

Note: 1952–1992 information is from Campbell (1996) using the first poll data in September. From 1996 to 2020, the "Incumbent Presidential Party Two-Party Vote (%)" column is based on the MIT Election Data and Science Lab (2017). "Early September Poll for Incumbent Party's Candidate" column is based on three sources: (1) 1996 to 2008: Gallup Presidential Election Trial-Heat Trends, 1936–2008 with Likely Voters (Gallup, 2023); (2) the American Presidency Project (Woolley & Peters, 2023); and (3) 2016 and 2020: RealClear Politics (RealClearPolitics, 2023a, 2023b). When the poll includes a minority party candidate, the poll percentage is divided evenly between the major parties. If possible, we use the first September for the results of the given year. For 2016 only, we used the average of three polls fielded during the first seven days in September (9/1/2016-9/7/2016) due to the multiple polls available in the same period, and they were heading in different directions. "Predicted Winner" is coded as *Right* if the incumbent party's candidate was winning (losing) against another candidate in the poll, and the incumbent party's candidate won (lost) the popular vote; otherwise coded as *Wrong*.

These final polls had a mean error of ±3.22. Further, they correctly called 14 of the 18 contests in terms of the incumbent party vote. These results appear close to the ANES tally of, respectively, 3.68 percentage points and 15 out of 18 contests won. However, the differences are noteworthy. While they both were in error with respect to 1960, they differed in others. In particular, examine the period from 2000: the final polls missed 2000, 2012, and 2016, while ANES missed only 2004. In other words, despite its hills and valleys, ANES has traditionally fared at least as well as leading commercial polls. If the forecasting focus falls on 21st-century contests, perhaps it has fared better.

9. Summary and conclusion

Our analysis demonstrates that aggregated post-election reported vote share accurately predicts aggregated actual vote share in the ANES data,1952–2020 (MAE of 2.23). We also demonstrate that aggregated pre-election intended vote share does not do too badly at forecasting aggregated actual vote share (MAE of 3.68). Moreover, our regression diagnostics reveal that accuracy is not significantly impaired by mode, weight, time, political party, or, perhaps most importantly, turnout. One obvious source of forecasting inaccuracy from the ANES vote intention questions comes from the fact that they occur a month or two before the election itself. Accuracy usually, but not always, improves the closer one gets to the election itself (Jennings et al., 2020; Lewis-Beck, 2005). Another argument is that the early measurement of the vote intention means the campaign effects are foregone. Still, the campaign effects themselves may be already 'baked into' the electorate's thinking by the summer of the election year (see, for example, Gelman and King, 1993; Holbrook, 1994; Lewis-Beck and Tien, 2018). Overall, the current study's findings are good news for forecasting researchers, especially considering that ANES surveys are not specifically designed to identify those who will or have voted in the past.

Nevertheless, the issues of sample size, normality, and outliers cannot be ignored. Depending on the test, we see that different observations may be troublesome (i.e., 1960, 1964, 1972, or 1980). However, there exists little consistency across tests (visual or mathematical). Thus, it becomes difficult to statistically single out a particular case as fatally flawed. Still, this does not give the data a "free pass" politically, in that for a given election, the forecast may be substantively way off. For example, in 1980, Reagan soundly defeated Carter in the two-party popular vote race, with a 55.2 percentage point to 44.8. The vote intention measure was off six percentage points, even giving Carter the popular vote win. On the one hand, from the perspective of forecasting, such an error

is discouraging. On the other hand, encouragement comes from examining the current pattern of errors from 2000 onward. Across these six elections, the ANES error never exceeds 3.9 and averages only 2.79 overall. (A further encouraging approach, which we do not pursue here, involves systematic combinations of forecasts that use different datasets or methodological approaches; on this, see Graefe et al., 2014).

One of the known limitations of using datasets from election studies is that the survey results typically are not released in time for them to be used in forecasting. In terms of ANES, they do have pre-election survey questionnaires – including the question of vote intention. If this preliminary dataset were available to the researchers, it could be used to "ex-ante" forecast on top of "ex-post" forecasts, such as those analyzed here. The institution/organization may not want researchers to use data as an "ex-ante" forecasting tool because it will naturally bring controversy, potentially leading to unfavorable consequences. Indeed, the ANES specifies that "The ANES design is focused on understanding behaviors and attitudes, rather than predicting election outcomes". Our findings suggest that the availability of these preliminary data would provide important resources that would be helpful to election forecasting. Still, we also see the potential disadvantages of the institution in doing so.

ANES's current negative lead time as a forecasting tool prevents its ex-ante use. However, another useful category of forecasting remains, namely, ex-post forecasting. As leading econometricians, Pindyck and Rubinfeld (1981, p. 204), noted early on: "It is useful to distinguish between two types of forecasting, *ex-post* and *ex-ante*. ... both forecasts predict values of a dependent variable beyond the time period in which the model is estimated. ... *ex-post* forecasts can be checked against existing data and provide a means of evaluating a forecasting model". This is something we do here, in our checking of the ex-post forecast against the existing data, to help evaluate the quality of the forecast (see also the useful discussion by statistician and forecaster Ledolter (2004, p. 397) on the definition of a forecast, which begins with "A forecast is a statement about an unknown and uncertain event—most often, but not always, a future event".).

Another methodological question is whether polls should be interpreted as forecasts at all – or whether a poll that will be used as a forecast should be weighted intentionally to reflect the outcome of the election. With regard to weighting the poll to match the electoral outcome once it is known, there are a number of logistical questions that the ANES would have to resolve before going down that road. While it is simple to understand the overall outcome, that is, the national vote count and vote counts by state and smaller jurisdictions, we do not know the demographics of voters because ballots are cast anonymously. So, we can weight the overall outcome to the correct numbers, but what about anything below that topline level? (See the following footnote for a more detailed discussion about these difficulties.[1])

[1] Most ANES analyses explain various subgroups and their differences. We could introduce errors at the subgroup level by weighting the topline outcome only.

Last but not least are the challenges coming from American electoral rules: the popular vote winner nationally does not necessarily indicate the winner in the Electoral College, as has happened twice in the last five elections (see the recent analysis by Hooghe et al. 2023). Further, as the ANES draws a national sample, its inferences are more confidently confined to the national popular vote (but see Murr 2015 for a possible exception).

Despite the turbulence of the 21st century, these popular contests have been rather well forecasted over that period via the ANES instrument. Consider the intended vote forecasts for two "turbulent" elections—2016 and 2020. For the former, the error was 3.30 points. For the latter, the error was just 2.22 points. This error is less than the average for the entire series (recall Table 3). Such precision appears especially impressive in light of the widespread error of the published pre-election polls going into these contests. As Jackson and Lewis–Beck Jackson and Lewis-Beck (2022, p. 139) observed, "Collectively, the polls performed worse than in 2016. The error in the 2020 pre-election polls at the national level was the highest in forty years". Seen in that light, the notion of the ANES as a gold standard retains its luster.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Weighting the subgroups to the actual outcome is not feasible at the same level of certainty that the rest of ANES weights operate. ANES weights are based on census-level data; we simply don't have data of that quality for voting subgroups. We can know from the state voter registration files that a person voted, but we will not know for whom; indeed, we will not know most of that individual's demographic information. While most voter files will have registrants' gender and age, not all will go further. Only a few states collect registrants' race, and while most states have voters affiliated with a party, there are 19 that do not. Due to these information gaps, we do not have a concrete source of information for how many people in various demographic groups voted. For example, if we want to know how many women voted for former President Donald Trump, even though their gender is on the voter registration lists, since we don't know for whom the women who voted cast their ballot, we cannot say for certain the number who voted for Trump. There are sources for subgroup estimates of the vote count, notably from the National Exit Poll, the AP-NORC VoteCast poll, and Catalist modeling off the voter file. However, the subgroup estimates from these sources often differ, and in the case of the exit poll and VoteCast, they are weighted to the topline election outcome. As a weighting tool, these estimates would be parameters generated by the same process we have already said would be problematic for ANES. The Catalist estimates offer more promise, as they are models from the voter file, so the starting point is the known population of voters. However, since the voter file information does not have complete demographic information or for whom ballots were cast – as mentioned above – these models are still functionally estimates, subject to the same error terms that all surveys have. Whether that error is low enough to warrant use as a weighting parameter is an open question. This analysis does not answer that question, as it is concerned with studying the existing error rates in ANES presidential election data.

Data availability

Replication materials can be found at https://github.com/hey-ikon/replication-materials/tree/main/IJF-2024-ANES.

References

Achen, C. H., & Blais, A. (2015). Intention to vote, reported vote and validated vote 1. In *The act of voting* (pp. 195–209). Routledge.

Aldrich, J. (2017). The American national elections study as "gold standard" for survey research in the twenty-first century.

American National Election Studies (2022). ANES 2020 time series study full release [dataset and documentation]. September 16, 2022 version. https://www.electionstudies.org.

Anderson, R. (2004). Regression diagnostics. In M. Lewis-Beck, A. E. Bryman, & T. F. Liao (Eds.), *Sage encyclopedia of social science research methods* (pp. 939–941). Sage Publications.

Ansolabehere, S., & Hersh, E. (2012). Validation: What big data reveal about survey misreporting and the real electorate. *Political Analysis*, *20*(4), 437–459.

Bernstein, R., Chadha, A., & Montjoy, R. (2001). Overreporting voting: Why it happens and why it matters. *Public Opinion Quarterly*, *65*(1), 22–44.

Bowen, B. D., & Weisberg, H. F. (1980). *An introduction to data analysis*. CA: WH Freeman San Francisco.

Campbell, J. E. (1996). Polls and votes: the trial-heat presidential election forecasting model, certainty, and political campaigns. *American Politics Quarterly*, *24*(4), 408–433.

Campbell, J. E. (2010). Explaining politics, not polls: Reexamining macropartisanship with recalibrated NES data. *Public Opinion Quarterly*, *74*(4), 616–642.

Campbell, J. E. (2016). Forecasting the 2016 American national elections. *49*, (4), (pp. 649–654).

Campbell, A., Converse, P. E., Miller, W. E., & Stokes, D. D. (1960). *The American voter*. University of Chicago Press.

Campbell, J. E., & Lewis-Beck, M. S. (2008). US presidential election forecasting: An introduction. *International Journal of Forecasting*, *24*(2), 189–192.

Clinton, H. R. (2017). *What happened*. Simon and Schuster.

Dassonneville, R., & Tien, C. (2021). Introduction to forecasting the 2020 us elections. *PS: Political Science & Politics*, *54*(1), 47–51.

Enamorado, T., & Imai, K. (2019). Validating self-reported turnout by linking public opinion surveys with administrative records. *Public Opinion Quarterly*, *83*(4), 723–748.

Fiorina, M. P. (1981). *Retrospective voting in American national elections*. New Haven: Yale University Press.

Gallup (2023). Gallup presidential election trial-heat trends, [1936–2008]. data retrieved from Gallup, https://news.gallup.com/poll/110548/Gallup-Presidential-Election-Trial-Heat-Trends.aspx.

Gelman, A., & King, G. (1993). Why are American presidential election campaign polls so variable when votes are so predictable? *British Journal of Political Science*, *23*(4), 409–451.

Goren, P. (2013). *On voter competence*. Oxford University Press.

Graefe, A., Armstrong, J. S., Jones, R. J., & Cuzán, A. G. (2014). Combining forecasts: An application to elections. *International Journal of Forecasting*, *30*(1), 43–54.

Holbrook, T. M. (1994). Campaigns, national conditions, and US presidential elections. *American Journal of Political Science*, 973–998.

Hooghe, M., Stiers, D., & Lewis-Beck, M. S. (2023). Has the electoral college grown more disproportional? An analysis of election results, 1876–2020. *Politics & Policy*, *51*(2), 167–183.

Jackman, S., & Spahn, B. (2019). Why does the American national election study overestimate voter turnout? *Political Analysis*, *27*(2), 193–207.

Jackson, N. (2018). The rise of poll aggregation and election forecasting. In L. R. Atkeson, & R. M. Alvarez (Eds.), *The oxford handbook of polling and survey methods*. Oxford University Press.

Jackson, N., & Lewis-Beck, M. S. (2022). Forecasting the party vote in the 2020 election. In J. C. Lucas, C. J. Galdieri, & T. S. Sisco (Eds.), *Polarization and political party factions in the 2020 election* (pp. 139–150). Rowman & Littlefield.

Jennings, W., Lewis-Beck, M., & Wlezien, C. (2020). Election forecasting: Too far out? *International Journal of Forecasting*, *36*(3), 949–962.

Lau, R. R., & Redlawsk, D. P. (1997). Voting correctly. *American Political Science Review*, *91*(3), 585–598.

Ledolter, J. (2004). Forecasting. In M. Lewis-Beck, A. E. Bryman, & T. F. Liao (Eds.), *The sage encyclopedia of social science research methods* (pp. 397–402). Sage Publications.

Lewis-Beck, M. S. (2005). Election forecasting: Principles and practice. *The British Journal of Politics and International Relations*, *7*(2), 145–164.

Lewis-Beck, C., & Lewis-Beck, M. (2015). Applied regression: An introduction. *vol. 22*, Sage publications.

Lewis-Beck, M. S., Norpoth, H., Jacoby, W. G., & Weisberg, H. F. (2008). *The American voter revisited*. University of Michigan Press.

Lewis-Beck, M. S., & Quinlan, S. (2019). The hillary hypotheses: testing candidate views of loss. *Perspectives on Politics*, *17*(3), 646–665.

Lewis-Beck, M. S., & Tien, C. (2018). Candidates and campaigns: How they alter election forecasts. *Electoral Studies*, *54*, 303–308.

Linzer, D., & Lewis-Beck, M. S. (2015). Forecasting US presidential elections: New approaches (an introduction). *International Journal of Forecasting*, *31*(3), 895–897.

McDonald, M. P. (2023). Voter turnout data. data retrieved from United States Elections Project, https://www.electproject.org/election-data/voter-turnout-data.

MIT Election Data and Science Lab (2017). U.S. president 1976–2020. http://dx.doi.org/10.7910/DVN/42MVDX, data retrieved from United States Elections Project.

Murr, A. E. (2015). The wisdom of crowds: Applying Condorcet's jury theorem to forecasting US presidential elections. *International Journal of Forecasting*, *31*(3), 916–929.

Nadeau, R., Dassonneville, R., Lewis-Beck, M. S., & Mongrain, P. (2020). Are election results more unpredictable? A forecasting test. *Political Science Research and Methods*, *8*(4), 764–771.

Petrocik, J. R. (2009). Measuring party support: Leaners are not independents. *Electoral Studies*, *28*(4), 562–572.

Pindyck, R. S., & Rubinfeld, D. L. (1981). *Econometric models and economic forecasts*. McGraw-Hill.

RealClearPolitics (2023a). RealClearPolitics - election 2016 - general election: Trump vs. clinton]. data retrieved from RealClearPolitics, https://www.realclearpolitics.com/epolls/2016/president/us/general_election_trump_vs_clinton-5491.html.

RealClearPolitics (2023b). RealClearPolitics - election 2020 - general election: Trump vs. biden]. data retrieved from RealClearPolitics, https://www.realclearpolitics.com/epolls/2020/president/us/general_election_trump_vs_biden-6247.html.

Stewart, M. C., Clarke, H. D., & Borges, W. (2019). Hillary's hypothesis about attitudes towards women and voting in the 2016 presidential election. *Electoral Studies*, *61*, Article 102034.

Traugott, E. C. (1989). On the rise of epistemic meanings in English: An example of subjectification in semantic change. *Language*, 31–55.

Weisberg, H. F., & Bowen, B. D. (1977). *An introduction to survey research and data analysis*. WH Freeman.

Woolley, J., & Peters, G. (2023). The American presidency project. data retrieved from The American Presidency Project, https://www.presidency.ucsb.edu/statistics/data.